

Chapter – 8 : Agile in Practice

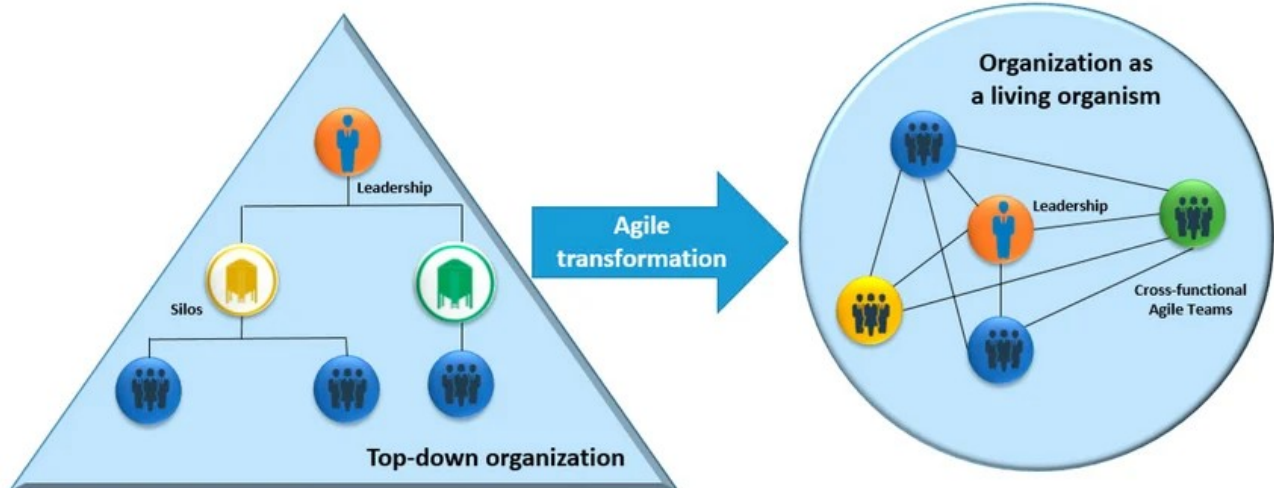
8.1. Case Studies of Agile Success and Failures Analyzing real-world examples

Assignment:

- Write 2 Agile success case studies from real companies/projects.
 - Mention: what was successful and why.
- Write 2 Agile failure case studies from real companies/projects.
 - Mention: what failed, why it failed, and what could have been improved.
- Use real examples and focus on Agile practices (teamwork, delivery, communication, etc.).

8.2. Agile Transformation - Steps to transition from traditional to Agile

- Agile Transformation is the organizational shift toward adopting Agile principles, practices, and culture to improve value delivery and collaboration.
- It is not just about:
 - Daily stand-ups
 - Using Jira
 - Running sprints
- It involves:
 - Mindset change
 - Structural and cultural change
 - New ways of leading, working, and learning



Why Organization Choose Agile Transformation:

- Faster delivery of products and services
- Better response to changing customer needs
- Improved team collaboration and communication
- Higher product quality through continuous feedback and testing
- Increased flexibility and adaptability to market changes
- Faster problem detection and continuous improvement

Steps to Transition from Traditional to Agile:

1. Create Urgency & Leadership Support
2. Define Agile Vision and Goals
3. Assess the Current State (As-Is Analysis)
4. Design the Target Agile Model
5. Form Agile Teams & Define Roles
6. Start with Pilot Projects
7. Scale Agile Across the Organization
8. Measure, Inspect, and Improve

Step 1 – Create Urgency & Leadership Support

- Explain why the organization needs Agile transformation.
- Highlight problems such as:
 - slow delivery,
 - customer dissatisfaction,
 - innovation delays,
 - delivery bottlenecks.
- Involve leadership and secure executive support for the transformation.
- “Without leadership support, Agile becomes theater.” - looks fake or surface-level Agile, not actually practiced
- Example: A company adopts Agile after customers complain about delayed software updates.

Step 2 – Define Agile Vision and Goals

- Clearly define what Agile success means for the organization.
- Common goals:
 - faster releases,
 - better product quality,
 - improved teamwork,
 - higher customer satisfaction.
- Example: Reduce software release time from 6 months to 1 month.

Step 3 – Assess the Current State (As-Is Analysis)

- Analyze the organization's current processes and culture:
 - team structures,
 - workflow,
 - decision-making,
 - tools and practices.
- Common assessment tools:
 - Agile Maturity Assessment:
 - Measures how “Agile-ready” an organization is
 - Example: A team uses Scrum meetings but still follows waterfall planning → low maturity
 - surveys:
 - Collect feedback from employees about current process.
 - Example: Developers report “too many approvals slow development”
 - interviews: deeper understanding than surveys
 - Value Stream Mapping:
 - Visual mapping of how work flows from idea → delivery.
 - Identifies delays and waste
- Example: Identifying that testing delays are slowing software releases.

Step 4 – Design the Target Agile Model

- Choose suitable Agile approaches:
 - Scrum: Iterative framework using sprints and defined roles
 - Kanban: Visual workflow system focused on continuous delivery
 - SAgile: Scaled Agile framework for large enterprises
 - LeSS: Lightweight scaling framework for multiple Scrum teams
 - Spotify Model: Squad-based (small teams) Agile model with autonomous teams and tribes (groups – working in similar area)

- Define:
 - Agile team structures,
 - roles,
 - governance : rules
 - metrics.
- Example: A large company adopts SAFe to coordinate multiple Agile teams.

Step 5 – Form Agile Teams & Define Roles

- Create cross-functional Agile teams with clear responsibilities.
 - team that has all required skills in one group to complete work without depending on other teams.
 - A mobile app team includes: 2 Developers (build app), 1 Tester (checks bugs), 1 UI/UX Designer (designs screens), 1 Scrum Master (facilitates work)
- Common roles:
 - Product Owner (PO),
 - Scrum Master (SM),
 - Developers,
 - Testers,
 - Designers.
- Avoid unclear or part-time roles.
- Example: A Scrum team includes developers, testers, and UI designers working together.

Step 6 – Start with Pilot Projects

- Begin Agile transformation with small pilot teams or projects.
- Provide:
 - training,
 - Agile tools,
 - coaching,
 - management support.
- Benefits of Pilots
 - Quick wins,
 - lower risk,
 - learning from experience,
 - building confidence.
- Example: One department first tests Scrum before company-wide adoption.

Step 7 – Scale Agile Across the Organization

- Expand successful Agile practices to more teams and departments.
- Support scaling with:
 - Agile coaches:
 - Experts who guide teams to correctly follow Agile practices
 - Communities of Practice (CoP):
 - Groups of people with the same skill who share knowledge across teams
 - continuous training:
 - Ongoing learning to improve Agile skills and tools
 - Eg. Regular workshops on Jira, Scrum, or testing techniques for all teams.
- Scaling Options
- SAFe → structured large organizations
- Spotify Model → innovative companies
- LeSS → product-focused organizations
- Example: Multiple development teams coordinate through Agile Release Trains (ARTs).

- An Agile Release Train (ART) is a group of multiple Agile teams that work together to deliver a large product.

Step 8 – Measure, Inspect, and Improve

- Continuously track Agile performance using metrics such as:
 - Cycle Time,
 - Lead Time,
 - Team Velocity,
 - Customer Feedback,
 - Employee Engagement.
- Conduct retrospectives and improve processes regularly.
- Agile transformation is a continuous improvement process, not a one-time change.
- Example: Teams improve sprint planning after identifying repeated delays during retrospectives.

Challenges in Agile Transformation:

1. Resistance to Change

- People often find it difficult to move from traditional ways to Agile practices.

From	Form of Restriction
Senior Management	Fear of Losing Control ; Leaders worry that Agile gives more decision power to teams
Middle Management	Threat to Authority; Managers feel their role becomes less important because teams become self-organizing
Teams	Comfort with known processes
Support Functions	“This doesn’t apply to us” mindset; Departments like HR, Finance, or Admin think Agile is only for developers, so they resist change.

- Solution: Explain benefits clearly and provide proper training and support.

2. Wrong Understanding of Agile

- Many teams think Agile means just using Scrum or working faster.
- Some focus only on tools and meetings instead of real Agile mindset.
- Solution: Teach Agile values like flexibility, collaboration, and continuous improvement.
 - Focus on mindset change over processes and tools

3. Unclear Roles and Responsibilities

- Confusion happens between roles like Scrum Master, Product Owner, and Project Manager.
- Some roles are misused, like Scrum Master becoming a reporter instead of a coach.
- Solution: Clearly define roles and give proper role-based training.
 - Leadership demonstrating correct Agile behaviors

4. Team Silos and Poor Collaboration

- Agile fails when teams work in isolation instead of collaboration.
- Silos (work separately / in isolation) reduce:
 - shared ownership
 - flow of value
 - feedback speed

- Solutions:
 - Build cross-functional teams (one team including all roles together ie UI, backend, testing)
 - Use shared KPIs
 - Conduct cross-team planning (e.g., Program Increment (PI) Planning)

5. Inconsistent Leadership Support

- Leaders may say they support Agile but still follow old management styles.
- This creates confusion and blocks Agile adoption.
- Solution: Train leaders in Agile thinking and encourage supportive leadership style.

8.3. Assignment/ Project - Applying Agile methodologies in a team project, Presentations and evaluations

Objective: Apply Agile methodologies, tools, communication, coordination, testing, and visualization techniques in a real team-based software project.

Project Task:

- Form a team
- select a mini project (You may implement on your ongoing minor project)
- actual software implementation (coding) is not required
- The objective is to understand and demonstrate the application of Agile principles and practices

Agile Requirements:

1. Product Backlog

- Create and maintain: User Stories (US), Tasks (TSK), Bugs (BUG)
- Include: Priority, Acceptance Criteria, Status

2. Sprint Planning and Execution

3. Coordination and Communication

4. Agile visualization techniques

5. Testing and quality practices

6. CI/CD and DevOps Practices

Submission Guidelines:

Create a document including all the above mentioned agile requirements and submit.

Include a title page with the project title and details of each team member.

The Project carries 10 marks towards your internal evaluation.

Submit document to brr.baral@gmail.com

Deadline: One week after course completion.